



Module 11 Day 3: Cryptography

Activity File: Cryptographic Attacks

You will continue your role as a security analyst working for the Hill Valley Police Department.

- Captain Strickland shared that they have been secretly capturing and monitoring all the online activity of the Hill Valley detectives.
- They believe that Detective Tannen is the Alphabet Bandit, as Tannen has been recently saving suspicious encrypted files on their computer.
- The captain believes these encrypted files contain evidence that can be used to prosecute Detective Tannen.
- While capturing Detective Tannen's online activity, Captain Strickland was able to capture the following:
 - Detective Tannen's password encryption script, which generates encrypted passwords from plaintext.
 - Detective Tannen's encrypted password, which is cbzhptmm.
- Your task is to see if you can figure out the algorithm of the encryption script to figure out Detective Tannen's plaintext password.
- Then, you will try to use Detective Tannen's password to log into their account and gather evidence that they are the Alphabet Bandit.

Instructions

1. Enter any plaintext into the Password Encrypter and determine the algorithm being used for encryption.
- [Detective Tannen's Password Encryption Script](#)
 - To run the password encrypter, run `python3 encrypter.py`.
Hint: Try multiple plaintext passwords to help determine the algorithm.
3. Once the algorithm has been determined for encryption, apply this method in reverse to determine Tannen's plaintext password from his encrypted password, cbzhptmm.
4. Apply this password on Tannen's login website to confirm that the password is correct. Look for any hard evidence that suggests Detective Tannen is the Alphabet Bandit.
 - To test Tannen's login password, run `python3 password.py`.

- [Detective Tannen's Computer Login](#)